Critical-Miss Evaluation in Small-Sample BEV

Log-Weight Sensitivity and Risk-Monitor Limits

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When do improved detection metrics translate into fewer critical misses, and do frozen risk scores help identify the remaining failures?

Exploratory nuScenes mini study: **404 frames from 8 capture logs**. The original A matrix retains 176/192 groups; complete gate comparisons use seeds 0/1. Seed 2 has a separate 243-frame, 6-log cohort. Risk monitoring uses three model replicas on the same 404 frames.

1 / Recall and frame failures

In A2/seed1/clean, pedestrian hits rise from **1,000/2,923 to 1,007/2,923**, yet all 17 count-improved frames retain a pedestrian miss. Joint-VRU AnyMiss stays **288/307**, with exactly the same failed-frame identities. Additional hits can leave a frame failed.

Local detection-metric disagreement

Separate P2 development case: fixed 8k updates, 41 frames, 1 log, 6 pedestrian instances/67 observations and 31 target frames. Other critical VRU classes have no support here.

Local P2 metric	Control	P2
mAP (%)	2.1758	2.5328
NDS (%)	7.2406	9.2326
Pedestrian AP (%)	0	0
Pedestrian / joint hits	8/67	15/67
AnyMiss / target frames	30/31	30/31

One frame recovers and one newly fails. Pedestrian outputs change from 2,175 to 2,177; all-range unmatched outputs from 2,165 to 2,160 (score >= 0.1). These are output-burden diagnostics, not official FP/precision. Aggregate AP gain does not improve pedestrian AP.

2 / Aggregation matters

Observation weighting and equal-log weighting can reverse an effect direction. In one A2/seed0/LiDAR comparison, pooled pedestrian recall changes by **-0.2053 percentage points**, while equal-log recall changes by **+0.0756 points**. These describe the same population under different weights.

3 / Ranking and alarm burden

The event is AnyMiss: at least one critical VRU missed in a frame, not collision risk. At **80/404 expected alarms (19.802%)**, captured failures are:

Seed	LR / failures	Dev	Conf	Random
0	57.58/270	40.49	47.00	53.47
1	43.56/272	39.51	46.00	53.86
2	51.47/266	36.54	35.00	52.67

LR exceeds random expectation only in seed 0; the frozen consistency criterion fails. Dev is same-fold development prevalence; Conf is one minus maximum pedestrian confidence. Fractional capture counts are uniform, label-independent tie expectations, not executed random alarms.

Development-fixed LR thresholds show limited internal enrichment, but alarm fractions drift to **40.97%/23.90%/39.49%**; unflagged failure ratios remain **64.85%/63.04%/62.32%** (seeds 0/1/2). Unflagged failure ratios divide expected counts. Original detector outputs and mAP/NDS/pedAP remain unchanged.

Boundaries of the evidence

Eight historically inspected logs, repeated instances, overlapping folds, low absolute AP, imperfect controls and one-log development fitting limit inference. External replication is absent. P1 failure preserves NO_PROMOTION; B2/B3 retain resource/support/implementation stops. Stable gate efficacy, calibrated safety risk and external generalization remain unestablished.

Next question / proposed, not executed

Can a frozen monitor retain failure enrichment and stable alarm cost on fresh capture logs? Predefine comparators, budget and stopping criteria; report per-log and residual failures. Re-splitting exposed mini data is not fresh validation. This is a proposed next study, not an executed result.

Contribution: a reproducible case analysis of established evaluation principles, with frozen compact inputs and complete result tables; no new estimator. Code development used AI assistance.

GitHub and compact reproduction | Full technical report and complete tables